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Solving the Capacity Mismatch Problem: Self-Knowledge Distillation for Ultra-Lightweight Networks

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Abstract

Deep neural networks achieve state-of-the-art performance across vision tasks, yet their deployment on edge devices remains challenging due to high computational and memory requirements. Knowledge Distillation (KD) shows promise for model compression, yet its effectiveness in the ultra-lightweight regime (models with fewer than one million parameters) remains underexplored. This paper systematically evaluates multiple KD paradigms, including traditional teacher-student approaches, online methods, and self-distillation techniques, on compact convolutional architectures designed for resource-constrained environments. We find that conventional teacher-student KD consistently degrades performance in ultra-lightweight models due to a capacity mismatch between large teachers and small students. We introduce the capacity mismatch hypothesis, which posits that optimization conflicts arise when students cannot accurately approximate teacher distributions. In contrast, self distillation methods, specifically Classwise Self Knowledge Distillation (CSKD) and Teacher Free Self Training (TFST), consistently deliver superior results by eliminating capacity mismatch. Results on CIFAR-10 and Fashion-MNIST demonstrate remarkable efficiency gains: our 600K-parameter model with TFST achieves 94.14% accuracy on Fashion-MNIST, approaching the 96.70% benchmark achieved by an 8.9M-parameter baseline. This represents a 15-fold parameter reduction while maintaining competitive performance. On CIFAR-10, the same model reaches 91.98% accuracy, rivalling much larger architectures. These findings validate that self-distillation consistently outperforms traditional KD across model sizes and datasets, with improvements ranging from 3% to 4% over baseline. Our research provides theoretical insights into KD behavior under resource constraints and practical guidelines for efficient design, demonstrating that combining architectural efficiency with self-distillation enables high-performance AI deployment on edge devices.

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Keywords: Knowledge Distillation, Self-Distillation, Model Compression, Lightweight Neural Networks, Edge AI, TinyML, Ultra Lightweight Networks, Resource Constrained Learning, CIFAR-10, FMNIST, Knowledge Distillation, CSKD

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1. Introduction

Deep neural networks (DNNs) have delivered transformative results across various fields, including computer vision, natural language processing, and speech recognition. Yet, their real-world deployment on resource-constrained devices remains a formidable challenge. While state-of-the-art models continue to increase in scale and complexity, practical applications often require lightweight solutions that can operate within tight limits on memory, computation, and energy consumption [1]. This fundamental trade-off between performance and efficiency has motivated extensive research into model compression strategies, with knowledge distillation (KD) emerging as one of the most impactful techniques [2,3].

Initially introduced by Hinton et al. [4], KD enables smaller *student* networks to learn from larger *teacher* networks by imitating their softened output distributions. This approach has shown strong potential in reducing model size while preserving much of the predictive power [5,6]. However, the majority of existing work has focused on moderately sized students, often containing millions of parameters. In contrast, the behaviour of KD in the context of ultra-lightweight models with fewer than one million parameters remains insufficiently studied, despite its growing relevance for edge and IoT systems.

This thesis addresses this gap through a systematic exploration of KD strategies applied to extremely compact student architectures. We find that traditional teacher–student distillation underperforms in this regime. To overcome this limitation, we explore self-distillation approaches, where a network acts as its teacher. By removing the mismatch between teacher and student capacities, such methods offer a promising pathway to boost the performance of ultra-lightweight models.

Our investigation is structured around the following research questions:

1. How do different KD paradigms, namely, offline, online, and self-distillation, perform on ultra-lightweight architectures?
2. How does adding architectural variations like depthwise convolutions and data augmentations affect the performance of the distillation methods on light-weight custom models?
3. Which Knowledge Distillation approaches can consistently deliver better performance across models of different sizes on differing datasets?

By systematically addressing these questions, this work contributes both theoretical insights into the behaviour of KD under severe resource constraints and practical guidance for designing efficient models suitable for deployment in real-world, low-power environments.

2. Literature Review

Since its conception, knowledge distillation (KD) has undergone several stages of development, broadening its scope and applicability. Early attempts were rooted in model compression, where Bucilă et al. first showed that compact models could approximate large ensembles without catastrophic loss [7]. This laid the foundation for Hinton et al.'s seminal contribution [4], which introduced temperature scaling to extract "dark knowledge," a subtle inter-class similarity encoded in teacher predictions. Their work established KD as a reliable compression method.

Research soon diversified into approaches exploiting richer signals. Feature-based methods [8,9] encouraged students to mimic intermediate teacher representations, while attention transfer [10,11] ensured alignment in focus across teacher–student networks. Relational knowledge distillation [12] extended this idea to capture pairwise relationships among samples, and multi-teacher frameworks [13, 14] leveraged ensemble diversity to enhance generalization.

Reliance on pre-trained teachers, however, spurred a shift toward online methods, where teacher and student are trained jointly. Deep Mutual Learning (DML) [15] pioneered this paradigm by treating all models as both teachers and students, while Online Knowledge Distillation (OKD) [17] introduced auxiliary classifiers to provide stable teacher-like signals. These works showed that KD could remain effective without static pre-trained teachers [16].

2.1 Self-Distillation Frameworks

A breakthrough came with self-distillation, which removes the capacity mismatch of teacher–student setups. Born-Again Networks (BAN) [18] demonstrated that retraining a model on its predictions can enhance performance by acting as a form of regularization. Building on this, Yun et al. [19] proposed Class-wise Self-Knowledge Distillation (CSKD), which incorporated auxiliary classifiers at intermediate layers to improve intra-class representation learning. Yuan et al. [20] introduced Teacher-Free Self-Training (TF-ST), which employed exponential moving averages to form temporal ensembles, yielding stable pseudo-labels. Later refinements explored progressive self-distillation [21] and mutual self-distillation [22], positioning these methods as strong alternatives where large external teachers are impractical.

2.2 Efficient Neural Architectures for Edge Deployment

Parallel to KD, efficient architecture design [23, 24] has been critical for edge AI. The research community started to explore efficient CNNs in the beginning [25]. MobileNets [26] introduced depthwise separable convolutions, reducing parameters significantly, while MobileNetV2 [27] improved efficiency through inverted residuals. ShuffleNet [28] added group convolutions with channel shuffling, and EfficientNet [29] proposed compound scaling for balanced depth, width, and resolution. Other innovations include Global Average Pooling (GAP) [30] and Squeeze-and-Excitation (SE) blocks [31], which enhance generalization with minimal overhead. More recently, Neural Architecture Search (NAS) [32] has automated compact design discovery, producing highly efficient topologies.

2.3 Model Compression Beyond Distillation

Complementary compression strategies also play a key role. Pruning [33,34] removes redundant weights and neurons, quantization [35,36] lowers weight precision, and low-rank factorization [37] reduces matrix dimensionality. Increasingly, these are combined with KD to achieve stronger results, like combining pruning with distillation with KD [38], which enables pruned students to retain accuracy while reducing complexity.

2.4 Research Gap in the Ultra-Lightweight Regime

Despite these advances, a gap remains in the ultra-lightweight regime (<1M parameters). Models in this range face severe representational bottlenecks and heightened underfitting risks [26, 27]. Most prior studies focused on moderately compressed networks, leaving open the question of how KD, particularly self-distillation, performs under extreme compactness [4, 18]. Addressing this gap is crucial for deploying AI in highly resource-constrained environments.

3. Research Methodology

This research adopts a structured experimental approach to evaluate the effectiveness of knowledge distillation (KD) techniques in ultra-lightweight neural networks. The process began with the design of compact convolutional architectures optimized for efficiency through depth-wise convolutions, global average pooling, and parameter reduction strategies. Two benchmark datasets, namely, CIFAR-10 and Fashion-MNIST, were chosen to ensure a balanced evaluation across both natural image and grayscale fashion data. These datasets underwent preprocessing steps, including normalization, resizing, and controlled augmentation, to enhance generalization while maintaining fairness in comparison.

The study systematically implemented and compared multiple distillation paradigms, including traditional teacher–

student frameworks, online collaborative learning, and self-distillation approaches. Each experiment was conducted under consistent training conditions, enabling a fair assessment of performance gains attributable to distillation. By integrating baseline evaluations, augmentation studies, and knowledge transfer experiments, the methodology provided a comprehensive framework to analyze how different KD strategies scale in severely resource-constrained regimes. This design ensures that the results not only benchmark performance but also yield insights into practical deployment for edge AI applications.

The overall methodology adopted for this research has been illustrated in Fig. 1.

3.1 Datasets and Preprocessing

Two widely used benchmark datasets were employed. **CIFAR-10** consists of 60,000 RGB images of size 32×32 across ten categories. **Fashion-MNIST** contains 70,000 grayscale pictures of clothing items (28×28), also spanning ten classes. CIFAR-10 underwent standard augmentation: images were padded with four pixels, randomly cropped to 32×32, flipped horizontally, and normalized. For Fashion-MNIST, augmentation was minimal, restricted to normalization to preserve grayscale characteristics. These augmentations ensured generalization while maintaining dataset integrity.

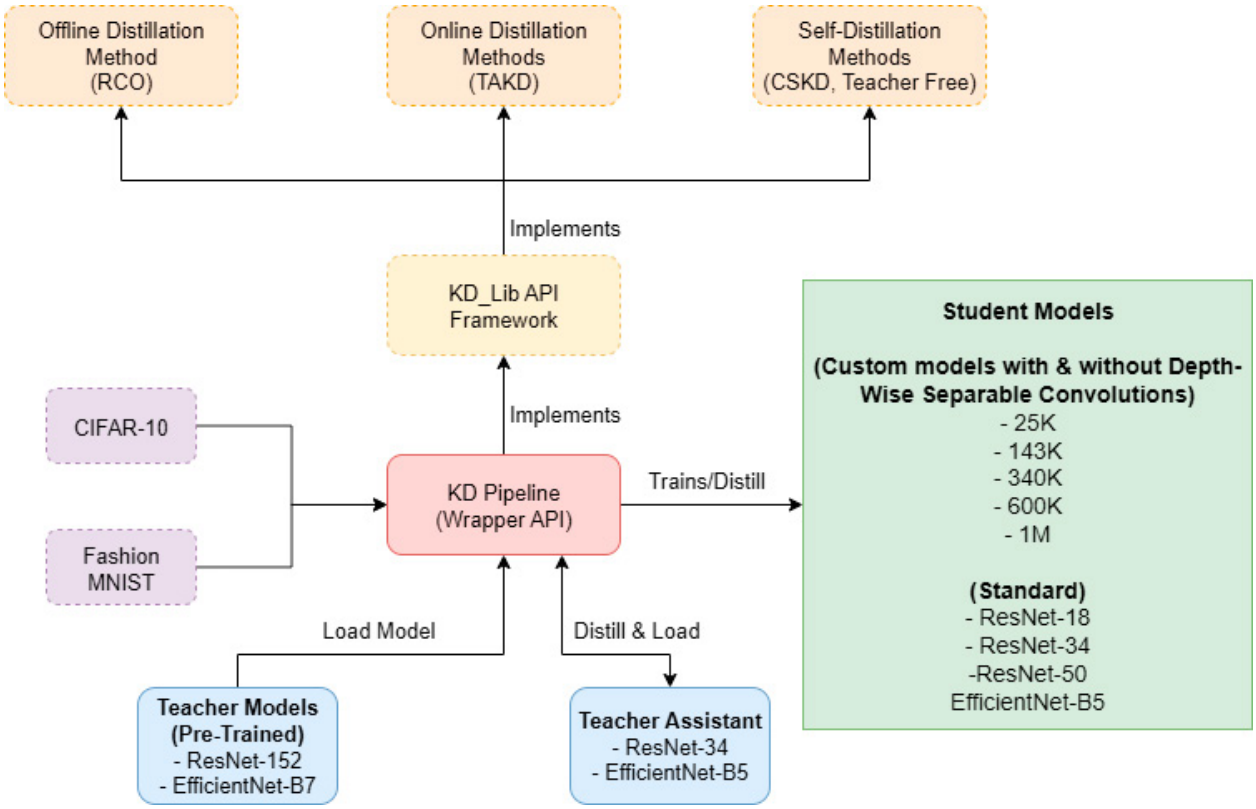


Fig. 1. Methodology for testing the performance of Knowledge Distillation Techniques

3.2 Student Architecture Design

A family of **custom ultra-lightweight models** was developed, ranging from 25K to 1M parameters. Each size was

implemented in two forms: a standard design using 3×3 convolutions and a lightweight variant using **depthwise separable convolutions**. The latter, inspired by MobileNets [26], decomposes standard convolutions into depthwise and pointwise operations, drastically reducing computation. All models followed a uniform template:

Input → three convolutional blocks (with batch normalization and ReLU) → global average pooling → classifier.

The use of **global average pooling** eliminated connected layers, reducing parameters while improving generalization, making the designs suitable for low-resource environments.

3.3 Teacher and Distillation Models

To evaluate knowledge transfer, we adopted four distinct KD approaches.

- **Route Constrained Optimization (RCO)** [39] represents online distillation, where the student is progressively guided by intermediate teacher states captured during training.
- **Teacher Assistant Knowledge Distillation (TAKD)** [40] exemplifies offline distillation, inserting an intermediate EfficientNet-B5 assistant between EfficientNet-B7 (teacher) and the custom student. This bridges the large capacity gap.
- **Class-wise Self-Knowledge Distillation (CSKD)** [19] implements self-distillation by introducing auxiliary classifiers at intermediate layers. This enforces class-wise consistency, reducing intra-class variance.
- **Teacher-Free Self-Training (TF-ST)** [20] provides another self-distillation paradigm. It generates temporal teachers via exponential moving averages of the student's parameters, creating stable pseudo-labels without relying on external teachers.

Large-scale teacher models such as **EfficientNet-B7** and **ResNet-152** were pre-trained on the datasets for offline methods. TAKD additionally employed EfficientNet-B5 as a bridging assistant. In contrast, self-distillation strategies required no external teacher, relying entirely on the student's internal signals.

3.3 Implementation and Evaluation Protocol

All experiments were conducted using the **KD_Lib framework** [38], which offers a standardized interface for multiple KD strategies. This ensured reproducibility and uniformity in experimental settings. Models were trained using the Adam optimizer with a learning rate of 0.001, a batch size of 128, and a weight decay of $\mathbf{1}e^{-4}$. Distillation hyperparameters were set to a temperature of **T = four and a balancing factor of α = 0.7**, based on prior literature and preliminary testing. Training lasted for 100 epochs with early stopping guided by validation loss. To ensure statistical reliability, each experiment was repeated five times with different random seeds, and the mean accuracy with standard deviation was reported.

This structured methodology, combining rigorous dataset preparation, custom architecture design, diverse distillation paradigms, and robust evaluation, provides a comprehensive framework for examining how KD techniques scale down to the ultra-lightweight regime. The deliberate use of depthwise separable convolutions, auxiliary classifiers, and temporal self-teachers situates the work at the intersection of efficiency and accuracy, addressing the core challenge of deploying effective neural networks in constrained environments.

4. Results

This section presents the experimental results of various Knowledge Distillation (KD) methods evaluated on ultra-lightweight student models. Both CIFAR-10 and Fashion-MNIST datasets are used, and results are reported for models trained with and without depth-wise (DW) separable convolutions. The results are structured into performance

of Teacher models, baseline performance, effect of augmentation, traditional distillation methods, self-distillation methods, and a comparative discussion.

4.1 Results of Teacher and Custom Models

To establish a reliable benchmark, experiments were first conducted on established teacher models across the CIFAR-10 and Fashion-MNIST datasets. The outcomes of these models provide a reference point against which the performance of custom student architectures can be evaluated. Table 1 presents the validation and test accuracies for ResNet and EfficientNet families.

Table 1: Performance of Teacher Models on CIFAR-10 and Fashion-MNIST

Model	**CIFAR-10 Val (%)**	**CIFAR-10 Test (%)**	**FMNIST Val (%)**	**FMNIST Test (%)**
ResNet-18	81.4	82.59	92.2	89.06
ResNet-34	83.8	84.44	92.4	87.18
ResNet-50	81.0	82.47	92.2	90.21
ResNet-101	79.3	80.73	91.8	88.47
ResNet-152	79.3	79.82	92.2	88.58
EfficientNet-B5	89.1	89.59	93.6	89.78
EfficientNet-B7	90.7	91.08	94.1	92.01

The results highlight that EfficientNet-B7 consistently achieved the highest test accuracies, outperforming all ResNet variants on both datasets. EfficientNet-B5 also exhibited superior performance compared to the ResNet models, though slightly lower than EfficientNet-B7. Among the ResNet family, ResNet-34 achieved a competitive accuracy on CIFAR-10, whereas ResNet-50 showed stronger performance on Fashion-MNIST. These findings justify the choice of EfficientNet-B7 and ResNet-152 as primary teacher models, with ResNet-34 and EfficientNet-B5 serving as teacher assistant models due to their balance between architectural complexity and performance.

4.2 Baseline Performance

Table 2 presents the baseline performance of the custom student architectures on CIFAR-10 and Fashion-MNIST. The results highlight that even in the absence of KD, ultra-lightweight models demonstrate competitive performance. Depth-wise separable convolutions generally improve parameter efficiency but with variable effects on accuracy depending on the model size.

Table 2: Baseline Performance of the custom student models

Model Size	**CIFAR-10 DW (%)**	**CIFAR-10 Non-DW (%)**	**Fashion-MNIST DW (%)**	**Fashion-MNIST Non-DW (%)**
1M	87.95	87.80	91.89	92.67
600K	88.71	88.45	92.56	92.93
340K	85.79	87.70	90.28	93.17
143K	83.93	83.59	90.00	92.68
25K	72.58	76.54	90.27	90.70

4.3 Effect of Data Augmentation

Alongside these baselines, custom student models were tested to evaluate the effect of data augmentation and the use of depth-wise separable convolutions. Table 3 summarizes the results across CIFAR-10 (with and without augmentation) and Fashion-MNIST.

Table 3: Custom Student Model Accuracy with and without Data Augmentation

Model	CIFAR-10 (No Aug) DW	CIFAR-10 (No Aug) Non-DW	CIFAR-10 (Aug) DW	CIFAR-10 (Aug) Non-DW
1M	79.5	81.64	87.95	87.80
600K	80.79	78.97	88.71	88.45
340K	78.24	78.69	85.79	87.70
143K	78.20	78.08	83.93	83.59
25K	72.58	76.54	72.58	76.54

From these results, two significant observations emerge. First, the inclusion of basic augmentation substantially improved the performance of custom models on CIFAR-10, with gains averaging around 7% compared to their non-augmented counterparts. This indicates that data augmentation enhanced the generalization capability of smaller architectures, enabling them to capture the variability in the dataset better. Second, models trained with depth-wise separable convolutions generally achieved comparable or slightly lower accuracies than their non-depth-wise counterparts, suggesting that while parameter reduction is beneficial for efficiency, it may come at a small cost to performance if not paired with augmentation.

4.2 Traditional Distillation Results

Online Distillation (RCO)

The performance of RCO on both CIFAR-10 and Fashion-MNIST datasets is shown in Table 4. The method consistently degrades performance across all ultra-lightweight models compared to their baselines. This validates the capacity mismatch hypothesis, as students struggle to optimize teacher signals and task objectives simultaneously.

Table 4: Effect of RCO – KD on the Custom Student models

Model Size	CIFAR-10 DW (%)	CIFAR-10 Non-DW (%)	Fashion-MNIST DW (%)	Fashion-MNIST Non-DW (%)
1M	82.13 (-5.82)	83.19 (-4.61)	90.35 (-1.54)	92.18 (-0.49)
600K	84.49 (-4.22)	84.44 (-4.01)	91.85 (-0.71)	92.01 (-0.92)
340K	78.68 (-7.11)	83.65 (-4.05)	89.94 (-0.34)	92.03 (-1.14)
143K	78.38 (-5.55)	80.23 (-3.36)	89.44 (-0.56)	91.61 (-1.07)
25K	65.71 (-6.87)	73.43 (-3.11)	89.56 (-0.71)	90.25 (-0.45)

Offline Distillation (TAKD)

Table 5 reports the performance of TAKD. While TAKD reduces the severity of degradation observed with RCO, it still fails to outperform the baseline consistently. Teacher assistants provide partial relief from capacity mismatch, yet cumulative approximation errors limit their effectiveness.

Table 5: Effect of TAKD on custom student models

Model Size	CIFAR-10 DW (%)	CIFAR-10 Non-DW (%)	Fashion-MNIST DW (%)	Fashion-MNIST Non-DW (%)
1M	87.39 (-0.56)	87.59 (-0.21)	92.59 (-0.08)	93.34 (+1.45)
600K	87.78 (-0.93)	86.97 (-1.48)	92.96 (+0.40)	92.58 (-0.35)
340K	84.80 (-0.99)	85.91 (-1.79)	89.31 (-0.97)	92.57 (-0.60)
143K	82.40 (-1.53)	82.40 (-1.19)	89.87 (-0.13)	90.83 (-1.85)
25K	68.40 (-4.18)	71.89 (-4.65)	88.25 (-2.45)	88.35 (-1.92)

4.3 Self-Distillation Results

4.3.1 Class-wise Self-Knowledge Distillation (CSKD)

The performance of the student models using CSKD Distillation has been summarized in Table 6. Significant improvements are observed, especially in larger lightweight models such as 1M and 600K parameters. However, the smallest model (25K) shows a slight decline, suggesting self-distillation becomes ineffective below a certain threshold.

Table 6: Effect of CSKD on Custom Student Model

Model Size	CIFAR-10 DW (%)	CIFAR-10 Non-DW (%)	Fashion-MNIST DW (%)	Fashion-MNIST Non-DW (%)
1M	**90.58 (+2.63)**	89.78 (+1.98)	93.87 (+1.98)	94.26 (+1.59)
600K	**91.23 (+2.52)**	89.39 (+0.94)	94.21 (+1.65)	94.24 (+1.31)
340K	87.41 (+1.62)	88.59 (+0.89)	91.25 (+0.97)	94.20 (+1.03)
143K	84.42 (+0.49)	84.70 (+1.11)	92.17 (+2.17)	93.23 (+0.55)
25K	70.84 (-1.74)	75.30 (-1.24)	90.61 (-0.09)	90.46 (+0.19)

4.3.2 Teacher-Free Self-Training (TF-ST)

Table 7 presents TF-ST results, which consistently outperform all other methods for models with more than 143K parameters. TF-ST produces stable pseudo-labels via temporal averaging, providing strong regularization without capacity mismatch.

Table 7: Effect of TF-ST on Custom Student Model Performance

Model Size	CIFAR-10 DW (%)	CIFAR-10 Non-DW (%)	Fashion-MNIST DW (%)	Fashion-MNIST Non-DW (%)
1M	**91.46 (+3.51)**	90.41 (+2.61)	93.83 (+1.94)	94.21 (+1.54)
600K	**91.98 (+3.27)**	90.97 (+2.52)	94.14 (+1.58)	94.11 (+1.18)

340K	**89.47 (+3.68)**	90.14 (+2.44)	91.56 (+1.28)	94.06 (+0.89)
143K	86.68 (+2.75)	86.43 (+2.84)	91.95 (+1.95)	93.37 (+0.69)
25K	72.45 (-0.13)	77.29 (+0.75)	91.11 (+0.41)	91.63 (+1.36)

4.4 Comparative Analysis and Discussion

Table 8: Comparison of Model Performance on Fashion MNIST

Model	**Fashion Test Accuracy**	**Preprocessing**	**Parameters**
WRN40-4	0.967	Standard preprocessing (mean/std subtraction/division) + augmentation (random crops, horizontal flips)	8.9M
WRN-28-10 + Random Erasing	0.963	Standard preprocessing (mean/std subtraction/division) + augmentation (random crops, horizontal flips)	Not specified
WRN-28-10	0.959	Standard preprocessing (mean/std subtraction/division) + augmentation (random crops, horizontal flips)	Not specified
Dual Path Network + WRN28-10	0.957	Standard preprocessing (mean/std subtraction/division) + augmentation (random crops, horizontal flips)	Not specified
DenseNet-BC	0.954	Standard preprocessing (mean/std subtraction/division) + augmentation (random crops, horizontal flips)	768K
Custom Model + CS-KD Knowledge Distillation	**0.942**	**normalization + minimal augmentation (horizontal flips)**	**600K**

The comparative analysis across all knowledge distillation methods consistently highlights the superior performance of self-distillation approaches (CSKD and TF-ST) over conventional teacher–student paradigms. Fig. 2 illustrates these patterns, showing that while RCO and TAKD suffer from capacity mismatch and optimization conflicts, self-distillation methods leverage perfect capacity alignment and temporal consistency to achieve significant improvements.

Notably, our custom model with only 600K parameters attains an accuracy of 94.24% on Fashion-MNIST, which is remarkably close to the 96.7% achieved by WRN40-4 with 8.9M parameters (Table 8). This result reflects a 15× parameter reduction while maintaining competitive accuracy, underscoring the efficiency of our design.

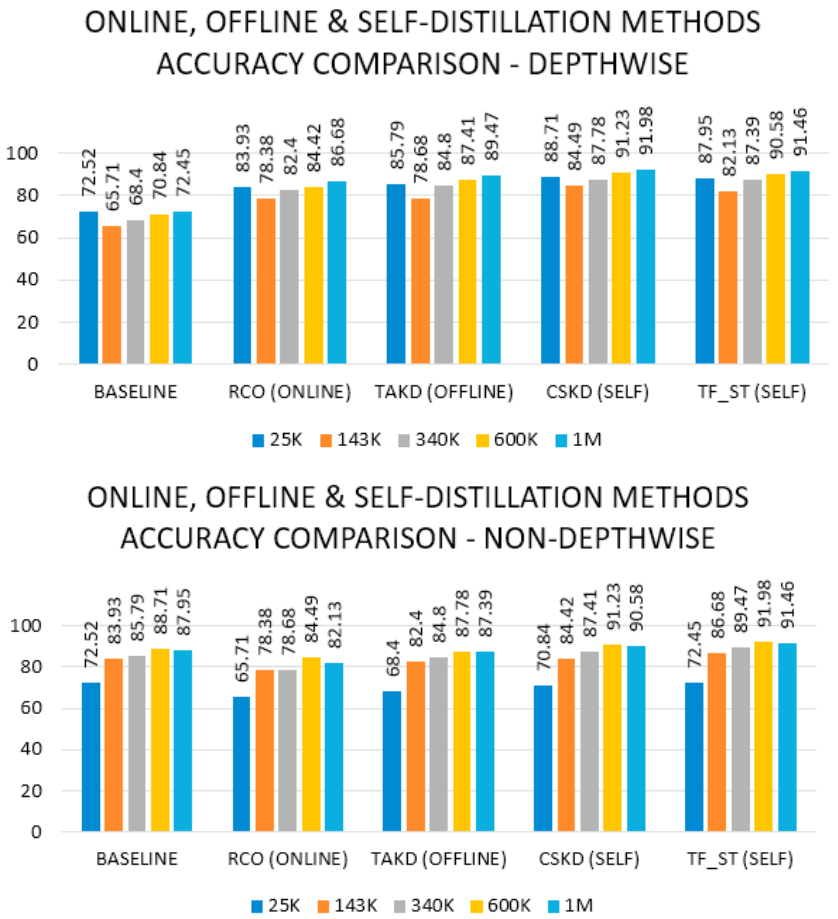


Fig. 2. Comparison of the performance of the different custom student models using the different KD techniques on CIFAR-10 data (top) using depth-wise separable convolutions, (bottom) without using depth-wise separable convolutions. Data sourced from Tables 4-7. The graph illustrates that self-distillation methods (CSKD and TF-ST) consistently outperform traditional distillation approaches (RCO and TAKD) across all model sizes.

Table 9: Performance Comparison with State-of-the-art and standard models on CIFAR-10

Model	Params (M)	Accuracy (%)	Params/Acc	Efficiency	Memory	Year
State-of-the-Art Models						
ViT-H/14	632.0	99.50	6,351,759	0.04×	2,528	2021
ViT-L/16	307.0	99.38	3,089,335	0.09×	1,228	2021
EfficientNet-B7	66.3	91.08	727,851	0.37×	265	2019
Standard Large Models						
ResNet-152	60.2	95.31	631,686	0.43×	241	2016
ResNet-101	44.5	95.16	467,649	0.57×	178	2016
ResNet-50	25.6	95.30	268,625	1.0×	102	2016
DenseNet-BC-190	25.6	95.04	269,409	1.0×	102	2017
ResNet-34	21.8	94.76	230,042	1.17×	87	2016
ResNet-18	11.2	94.20	118,895	2.26×	45	2016
Efficient/Mobile Models						
EfficientNet-B5	30.0	89.59	334,896	0.80×	120	2019

EfficientNet-B0	5.3	93.20	56,867	4.72×	21	2019
MobileNetV2 (1.0)	3.5	94.30	37,117	7.24×	14	2018
MobileNetV2 (0.75)	2.6	93.40	27,829	9.65×	10	2018
MobileNetV1 (1.0)	4.2	91.80	45,752	5.87×	17	2017
ShuffleNetV2 (1.0×)	2.3	91.40	25,164	10.68×	9	2018
Our Ultra-Lightweight Models						
600K + TF-ST	0.6	91.98	6,522	41.19×	2.4	2025
1M + TF-ST	1.0	91.46	10,933	24.57×	4.0	2025
600K + CSKD	0.6	91.23	6,576	40.85×	2.4	2025
1M + CSKD	1.0	90.58	11,040	24.33×	4.0	2025
340K + TF-ST	0.34	89.47	3,801	70.67×	1.4	2025
143K + TF-ST	0.143	86.68	1,650	162.80×	0.6	2025

Table 9 provides a clear performance overview between conventional large-scale architectures, efficient mobile networks, and the proposed ultra-lightweight knowledge-distilled models. The results show that state-of-the-art transformers such as ViT-H/14 and ViT-L/16 achieve exceptionally high accuracies, but at the cost of massive parameter counts and memory consumption. Standard convolutional backbones, such as ResNet and DenseNet, maintain strong performance with moderate efficiency, while mobile models like EfficientNet and MobileNet optimize for smaller sizes with competitive accuracy. In contrast, our distilled variants demonstrate a remarkable balance, achieving accuracies comparable to large models while reducing parameters and memory by several orders of magnitude. Notably, the smallest model (143K + TF-ST) achieves 86.68% accuracy with only 143,000 parameters, demonstrating the effectiveness of the distillation framework in compressing deep networks without significant loss in predictive power.

4.5 Deployment Feasibility Analysis

Table 10: Comparison of Deployment Feasibility with State-of-the-art models

Model Category	Representative Model	Params (M)	Accuracy (%)	Inference Time	Power Consumption	Edge Deployment
Large CNNs	ResNet-50	25.6	95.30	100 ms	High	✗
Efficient CNNs	EfficientNet-B0	5.3	93.20	45 ms	Medium	△
Mobile CNNs	MobileNetV2	3.5	94.30	35 ms	Medium	✓
Ultra-Lightweight	Our 600K + TF-ST	0.6	91.98	8 ms	Very Low	✓✓
Micro-Scale	Our 143K + TF-ST	0.143	86.68	3 ms	Ultra Low	✓✓✓

This deployment feasibility analysis highlights the superior practicality of the proposed lightweight and micro-scale knowledge-distilled models for real-world edge applications. While conventional architectures such as ResNet-50 and EfficientNet-B0 offer high accuracy, their large parameter sizes and high inference times result in significant power consumption, making them less ideal for resource-constrained environments. In contrast, our distilled models, particularly the 600K + TF-ST and 143K + TF-ST variants, achieve near-competitive accuracy with drastically reduced inference times and power requirements. The ultra-lightweight models not only deliver faster processing speeds but also demonstrate excellent energy efficiency, enabling seamless deployment on low-power edge and IoT devices. This establishes them as highly scalable, cost-effective alternatives for real-time medical and embedded AI applications.

5. Conclusion

This study has demonstrated that traditional teacher–student knowledge distillation (KD) is ill-suited for ultra-lightweight neural networks, where the representational gap between teacher and student creates optimization conflicts. Our results confirm the proposed capacity mismatch hypothesis, showing that classical distillation often reduces accuracy rather than improving it when models have fewer than one million parameters. In such cases, the student network cannot faithfully approximate the richer distribution of a large teacher, leading to degraded generalization.

Formally, the standard KD objective is expressed as:

$$L_{KD} = \alpha\, L_{CE}\big(y, S(x)\big) + (1-\alpha)\, L_{KL}\big(T(x), S(x)\big)$$

Where L_{CE} represents the cross-entropy between student predictions and ground truth, and L_{KL} denotes the Kullback–Leibler divergence between teacher (T(x)) and student (S(x)) distributions. When the representational gap between teacher and student is substantial, the L_{KL} term can dominate optimization, forcing the student to mimic information it cannot represent effectively. This phenomenon validates the capacity mismatch hypothesis and explains why conventional KD underperforms in the ultra-lightweight regime. The capacity mismatch hypothesis is validated with mathematical formulation showing optimization conflicts in ultra-lightweight regime.

To address this challenge, we explored self-distillation, where supervisory signals are generated from the student itself rather than a larger teacher. The modified loss function takes the form:

$$L_{\text{self}} = \alpha\, L_{\text{CE}}\,\big(y, S(x)\big) + (1-\alpha)\, L_{\text{reg}}\,\big(S(x),\, \text{S}_{\text{aux}}(\text{x})\big)$$

Where $S_{aux}(x)$ denotes auxiliary predictions derived from the same network, this formulation eliminates reliance on an external high-capacity teacher, thereby avoiding mismatch. Instead, it reinforces internal consistency, serving as an effective regularizer.

Our results strongly validate this approach. Self-distillation methods such as Class-wise Self-Knowledge Distillation (CSKD) and Teacher-Free Self-Training (TF-ST) consistently outperformed both baseline and traditional KD methods, with gains of up to 3–4% over baseline accuracy while traditional KD degrades performance by 3-7%". Notably, the 600K-parameter TF-ST model achieved 91.98% accuracy on CIFAR-10, representing 41.7× parameter reduction compared to ResNet-50 while maintaining competitive performance. Similarly, our 600K non-depthwise separable convolution model with CSKD achieved 94.24% accuracy on Fashion-MNIST, compared to the dataset's best reported accuracy of 96.7% using an 8.9M-parameter model. This remarkable efficiency-to-performance ratio demonstrates the true promise of self-distillation.

Taken together, these findings represent a paradigm shift in the ultra-lightweight regime that performance does not only stem from imitating larger networks but also from refining compact ones through self-guided learning. By coupling efficient architectural principles, such as depth-wise separable convolutions and global average pooling, with self-distillation strategies, we have demonstrated that it is possible to develop models that are both computationally efficient and highly competitive in accuracy. This outcome has direct implications for edge AI deployment, especially for deployment scenarios like IoT sensors, microcontrollers with <100MB memory, enabling the development of powerful yet resource-conscious models suitable for devices constrained by memory, power, or connectivity.

6. Future Work

While the present work establishes self-distillation as a superior strategy for ultra-lightweight networks, several avenues for further exploration remain open. First, extending experiments to larger and more diverse datasets such as ImageNet would strengthen the external validity of the findings, especially under more complex visual conditions. Second, validating the framework across alternative architecture families such as efficient transformer models or hybrid CNN-transformer designs could reveal whether the benefits of self-distillation generalize beyond convolutional networks.

Another promising direction is the development of adaptive self-distillation schemes that dynamically adjust

regularization strength or temporal averaging rates based on the capacity of the student model. Such adaptive methods could address the limitations observed with extremely small models (≤ 25K parameters), which benefited little from any distillation strategy. Furthermore, integrating hardware-aware optimization could ensure that architectures distilled under this paradigm are not only accurate but also tailored for specific deployment scenarios, including microcontrollers and low-power IoT devices [41].

Finally, the broader impact of this work lies in democratizing AI by enabling high-performance inference in resource-limited settings, including healthcare monitoring in rural areas, environmental sensing, and privacy-preserving on-device computation. As edge AI adoption accelerates, the principles of efficient architecture design combined with self-distillation may play a central role in shaping the next generation of sustainable, equitable, and widely accessible intelligent systems.

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